

Hao Wang 王豪

Leave the world a little different.

Ph.D. Candidate · Open-Ended Visual Perception · Multimodal Foundation Models · Agentic AI

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HCP Lab, Sun Yat-sen University & Pengcheng Laboratory
Open to industry research roles and research collaborations.

Expected graduation: **December 2026**

Profile

Ph.D. candidate whose research centers on **open-ended computer vision** and **multimodal large language models**, and who is increasingly exploring **multimodal agentic models**. First-author work includes X2SAM (ECCV 2026), X-SAM (AAAI 2026), OV-DINO, and TMANet (ICIP 2021), spanning unified image/video segmentation, open-vocabulary detection, and temporal pixel-level understanding. All representative works are open-sourced with project pages and code (1.1k+ GitHub stars in total). Interested in connecting foundation models with reliable dense perception and agentic workflows for practical vision systems.

Research Highlights

- Built a first-author research line on open-ended perception, from video semantic segmentation to open-vocabulary detection and multimodal any-segmentation.
- Developed multimodal segmentation MLLMs that accept both natural-language instructions and visual prompts, bridging high-level reasoning and dense mask prediction.
- Extended any-segmentation from images to videos with a mask-memory mechanism for temporally consistent object- and pixel-level understanding.
- Maintain open-source research code and an agent-oriented research toolchain (STAR / STORY / STAGE), emphasizing reproducible and open research practice.

First-Author Publications

X2SAM: Any Segmentation in Images and Videos

ECCV 2026

Hao Wang, Limeng Qiao, Chi Zhang, Guanglu Wan, Lin Ma, Xiangyuan Lan, Xiaodan Liang

A unified segmentation MLLM that extends any-segmentation from images to videos, supporting conversational instructions and visual prompts through Mask Memory for temporally consistent pixel-level perception. [Project](#) | [Paper](#) | [Code](#)

X-SAM: From Segment Anything to Any Segmentation

AAAI 2026

Hao Wang, Limeng Qiao, Zequn Jie, Zhijian Huang, Chengjian Feng, Qingfang Zheng, Lin Ma, Xiangyuan Lan, Xiaodan Liang

A unified MLLM framework that extends segment-anything capabilities to any segmentation, enabling instruction-driven interaction and pixel-level perceptual understanding. [Project](#) | [Paper](#) | [Code](#)

OV-DINO: Unified Open-Vocabulary Detection with Language-Aware Selective Fusion

arXiv 2024

Hao Wang, Pengzhen Ren, Zequn Jie, Xiao Dong, Chengjian Feng, Yinlong Qian, Lin Ma, Dongmei Jiang, Yaowei Wang, Xiangyuan Lan, Xiaodan Liang

A unified open-vocabulary detector pre-trained on diverse large-scale datasets with language-aware selective fusion, improving category generalization through visual-language alignment. [Project](#) | [Paper](#) | [Code](#)

TMANet: Temporal Memory Attention for Video Semantic Segmentation

ICIP 2021

Hao Wang, Weining Wang, Jing Liu

A temporal memory attention method for long-range video semantic segmentation that captures cross-frame relations without optical-flow prediction. [Paper](#) | [Code](#)

Open-Source Projects

Research code and tooling on GitHub; star counts as of September 2026.

OV-DINO arXiv 2024	★ 414	STAR Harness · WIP	★ 51
Unified open-vocabulary detection with language-aware selective fusion.		Systematic Toolchain for AI Research: an agent harness for planning, running, and tracking research.	
X-SAM AAAI 2026	★ 391	STORY / STAGE Harness · WIP	
From segment anything to any segmentation.		Companion toolchains for organizing research over years and for authoring, guiding, and editing.	
TMANet ICIP 2021	★ 128	omdsh-plugins Collection	
Temporal memory attention for video semantic segmentation.		Oh My DeepSeek Harness: a curated plugin collection for the DeepSeek Harness.	
X2SAM ECCV 2026	★ 104	arXivTeX Template	★ 33
Any segmentation in images and videos with one unified MLLM.		Paper template for arXiv or any conference. Also: Claude Model Proxy, Codex Remote Connector.	

Research Experience

Meituan · M17-MM	2025.01 – Present
Research Intern	
- Led X2SAM, a unified any-segmentation MLLM for images and videos with conversational instructions and visual prompts. - Designed a mask-memory mechanism that preserves object-level consistency in video segmentation and interactive perception; owned training, evaluation, and open-source release.	
Meituan · Vision Intelligence Department	2022.07 – 2025.01
Research Intern	
- Led OV-DINO and X-SAM, advancing open-vocabulary recognition and any segmentation. - Studied language-aware fusion and multimodal supervision for scalable visual recognition and segmentation.	
Tencent · AI Platform Department	2021.05 – 2021.08
Applied Research Intern	
Huawei · Photo Processing Department	2019.09 – 2020.07
Applied Project Intern	

Education

Sun Yat-sen University & Pengcheng Laboratory	2022.09 – Present
Ph.D. student, School of Intelligent Systems Engineering	
- Co-advised by Prof. Xiaodan Liang and Assoc. Prof. Xiangyuan Lan. - Research: multimodal foundation models, open-ended visual understanding, image/video segmentation, and agentic AI. Expected graduation: December 2026.	
University of Chinese Academy of Sciences & Institute of Automation, CAS	2019.09 – 2022.06
Master student, School of Artificial Intelligence; advised by Prof. Jing Liu	
Beijing Jiaotong University	2015.09 – 2019.06
Bachelor student, School of Electronic and Information Engineering	

Co-Authored Publication

WL-MSR: Watch and Listen for Multimodal Subtitle Recognition	ICASSP 2023
Jiawei Liu, Hao Wang , Weining Wang, Xingjian He, Jing Liu	
Transformer-based multimodal subtitle recognition that fuses OCR and ASR information with mask/crop strategies and multi-level identity embeddings.	

Service & Awards

Conference Reviewer	AAAI 2026, ECCV 2024, ICCV 2023
Journal Reviewer	Proceedings of the IEEE
Award	1st place, the 1st VSPW Challenge Workshop, ICCV 2021 (2021.09)